1. Fourier series in trigonometric form:

$$\boxed{x_p(t) = a_0 + 2\cdot\sum_{k=1}^{\infty}\left(a_k\cos\left(\frac{2\pi k}{T_p}t\right) + b_k\sin\left(\frac{2\pi k}{T_p}t\right)\right)}$$

where $a_k = (c_{-k}+c_k)/2$ and $b_k = (c_{-k}-c_k)/2$

2. Spectral properties of a real signal:

1) $x(t)$ is real, we have $X^*(f) = X(-f)$, which leads to: $|X(f)| = |X(-f)|$ and $\angle X(f) = -\angle X(-f)$
2) $x(t)$ is real and even, we have $X(f)$ is also real and even: $X^*(f) = X(f)$ and $X(f) = X(-f)$
3) $x(t)$ is real and odd, we have $X(f)$ is imaginary and odd: $X^*(f) = -X(f)$ and $X(f) = X(-f)$

3. Properties of the Dirac-δ function:

1) Symmetry: $\delta(t) = \delta(-t)$
2) Sampling: $x(t)\delta(t-\lambda) = x(\lambda)\delta(t-\lambda)$
3) Sifting: $\int_{-\infty}^{\infty} x(t)\delta(t-\lambda)dt = x(\lambda)$
4) Replication: $x(t) * \delta(t-\xi) = x(t-\xi)$
5) White spectrum: $\mathcal{F}\{\delta(t)\} = \mathcal{F}^{-1}\{\delta(t)\} = 1$

4. Use Fourier transform on periodic signals:

$$\boxed{X_p(f) = \sum_{k=-\infty}^{\infty} c_k\mathcal{F}\{e^{j2\pi k f_p t}\}}$$

$$\boxed{= \sum_{k=-\infty}^{\infty} c_k\cdot\delta(f-kf_p)}$$

5. Fourier transform of the generating function:

$x_p(t) = g(t) * \sum_n \delta(t-nT_p) = \sum_n g(t-nT_p)$
$\rightleftharpoons X_p(f) = \sum_k f_p G(kf_p)\delta(f-kf_p)$ thus $c_k = f_p G(kf_p)$

6. DC value (or called average value):

1) Original definition: $c_0 = \lim_{\tau\to\infty}\frac{1}{\tau}\int_{-\tau/2}^{\tau/2} x(t)dt = \frac{X(0)}{\infty}$.
2) For periodic signal: $c_0 = \frac{1}{T_p}\int_{-0.5T_p}^{0.5T_p} x(t)dt$.
3) In the frequency domain: $\frac{X(0)}{\infty} = \frac{\int_{-\infty}^{\infty} x(t)dt}{\infty} = c_0$.
4) The DC value of a signal is K if $X(f)$ contains a $K\delta(f)$ term.
5) All bounded energy signals have zero DC value.
6) All bounded periodic signals are power signals.

7. Energy & energy spectral density:

$$\boxed{E = \int_{-\infty}^{\infty}|x(t)|^2dt = \int_{-\infty}^{\infty}|X(f)|^2df = \int_{-\infty}^{\infty}E_x(f)df}$$

8. Power & power spectral density:

$P = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}|x(t)|^2dt = \int_{-\infty}^{\infty}\lim_{T\to\infty}\frac{1}{2T}|X_T(f)|^2df = \int_{-\infty}^{\infty}P_x(f)df$

Furthermore, for periodic signals, we have:
$P_x(f) = \sum_k |c_k|^2\delta(f-kf_p)$ and $P = \sum_k |c_k|^2$
c_k is fourier series coef

9. Magnitude & phase of a complex number:

1) Any complex number can be expressed as $X = a+jb = |X|\cdot e^{j\angle X}$,
whose magnitude is $|X| = \sqrt{X\cdot X^*} = \sqrt{a^2+b^2}$, and whose phase can be written as $\tan\angle X = \frac{\mathrm{Im}\{X\}}{\mathrm{Re}\{X\}} = \frac{b}{a}$.
2) In the form of $\frac{1}{a+jb}$, its magnitude is $\frac{1}{\sqrt{a^2+b^2}}$, and its phase is $-\tan^{-1}(\frac{b}{a})$.

10. Properties of an LTI (linear time-invariant) system:

1) Impulse response (in t-domain): $y(t) = x(t) * h(t)$.
2) Frequency response (in f-domain): $Y(f) = X(f)\cdot H(f)$.
3) With transfer function $\tilde{H}(j\omega)$, if input is $x(t) = A\cos(\omega t+\varphi)$, we have output $y(t) = A|\tilde{H}(j\omega)|\cos(\omega t+\varphi+\angle\tilde{H}(j\omega))$.

11. Stability of an LTI system with poles and/or zeros:

1) BIBO stable: All poles lie on the left-half s-plane, i.e. $\mathrm{Re}[s] < 0$.
2) Marginally stable: One or more poles lie on the $j\omega$ axis, the rest on the left-half s-plane.
3) Unstable: One or more poles on the right-half s-plane, or one or more repeated poles on the $j\omega$ axis.

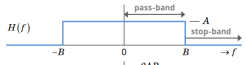
12. LTI Systems & Stability

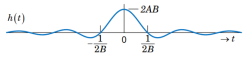
Properties: $y(t) = x(t) * h(t) \leftrightarrow Y(f) = X(f)\cdot H(f)$.
Steady-State Sinusoidal: If input $x(t) = A\cos(\omega_0 t+\phi)$, output $y(t) = A|\tilde{H}(j\omega_0)|\cos(\omega_0 t+\phi+\angle\tilde{H}(j\omega_0))$. Similarly, $Ae^{j(\omega_0 t+\psi)} \to A\tilde{H}(j\omega_0)e^{j(\omega_0 t+\psi)}$.
Transfer Fn: $\tilde{H}(s) = \frac{\sum_{m=0}^{M} b_m s^m}{\sum_{n=0}^{N} a_n s^n} = K\frac{\prod(s/z_m+1)}{\prod(s/p_n+1)}$. Roots p_n are poles, z_m are zeros. Pole-zero excess $= N-M$. Zeros affect transience/shape, NOT stability.
Stability: Depends entirely on poles $\mathrm{Re}(p)$.
1) BIBO: All $\mathrm{Re}(p) < 0$ (Left half s-plane).
2) Marginal: $\mathrm{Re}(p) \leqslant 0$, poles on imaginary axis $\mathrm{Re}(p) = 0$ are distinct. None on right half.
3) Unstable: Any $\mathrm{Re}(p) > 0$ (Right half) OR repeated poles on $\mathrm{Re}(p) = 0$.
Validity of $\tilde{H}(j\omega)$: $\tilde{H}(j\omega)|_{\omega=2\pi f} = H(f)$ everywhere if BIBO stable. If marginally stable, valid everywhere EXCEPT at the pole frequencies ω_o. (e.g., $1/s$ invalid at $\omega = 0$, $\sin(\omega_o t)u(t)$ invalid at $\omega = \pm\omega_o$).
DC Gain: $\tilde{H}(0)$. Integrators $\to\infty$, Diff $\to 0$. Don't express in dB.

13. Filters & Sampling

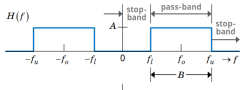
Total phase is (Phase of Numerator) - (Phase of Denominator). Two arctangent functions.
Ideal LPF: $H(f) = K$ for $|f| < f_c$, else 0. $H(f) = A\mathrm{rect}(\frac{f}{2B}) \leftrightarrow h(t) = 2AB\mathrm{sinc}(2Bt)$. Cutoff=BW=$B$.
Ideal HPF: $H(f) = K$ for $|f| > f_c$, else 0.
Ideal BPF: $H(f) = K$ for $f_1 < |f| < f_2$, else 0. $H(f) = A[\mathrm{rect}(\frac{f+f_o}{B}) + \mathrm{rect}(\frac{f-f_o}{B})] \leftrightarrow h(t) = 2AB\mathrm{sinc}(Bt)\cos(2\pi f_o t)$. Center $f_o = 0.5(f_u+f_l)$. BW$=B = f_u - f_l$.
Ideal Notch: $H(f) = 0$ for $f_1 < |f| < f_2$, else K.

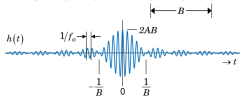
Ideal LPF





Ideal BPF



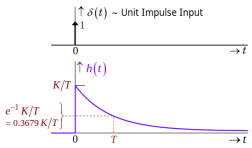


Reconstruction: Perfect signal requires gain $A = T_s = 1/f_s$. Ideal B is Nyquist Freq $f_s/2$. Division-by-2 needed for perfectly-overlapping BPF images.
Sampling: Nyquist rate $f_s \geqslant 2f_c + B$.
1) Perfectly-overlapping images: $f_s = 2f_c/k$, $k = 1, 2, \ldots, \lfloor 2f_c/B\rfloor$.
2) Un-aliased: $\frac{2f_c+B}{k+1} \leqslant f_s \leqslant \frac{2f_c-B}{k}$, $k = 1, 2, \ldots, \lfloor\frac{2f_c-B}{2B}\rfloor$.

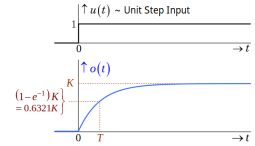
14. First & Second Order Systems

1st Order DE: $T\dot{y} + y = Kx \to \tilde{H}(s) = \frac{K}{Ts+1}$. Pole at $-1/T$.
Impulse $h(t) = \frac{K}{T}e^{-t/T}u(t)$. Step $o(t) = K(1-e^{-t/T})u(t)$.

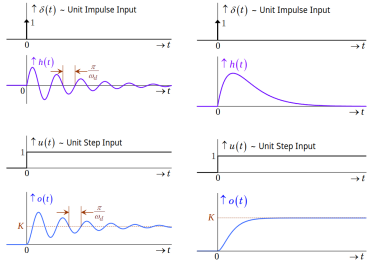
1st Order Impulse



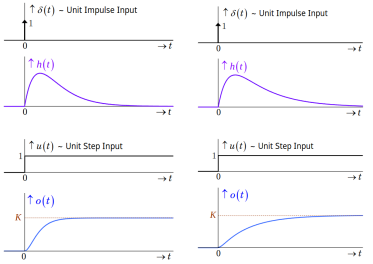
1st Order Step



2nd Order DE: $\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = K\omega_n^2 x \to \tilde{H}(s) = \frac{K\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$. Poles: $s_{1,2} = -\omega_n\zeta \pm \omega_n\sqrt{\zeta^2-1}$.
1) Overdamped ($\zeta > 1$): Real/distinct poles.
$h(t) = [K_1 e^{-p_1 t} + K_2 e^{-p_2 t}]u(t)$
$o(t) = K[1 - \frac{K_1}{p_1}e^{-p_1 t} - \frac{K_2}{p_2}e^{-p_2 t}]u(t)$
$s_{1,2} = -\omega_n\zeta \pm \omega_n(\zeta^2-1)^{1/2}$

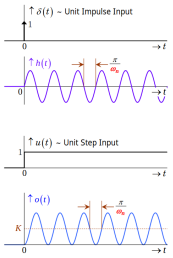


2) Critically damped ($\zeta = 1$): Real/repeated poles.
$h(t) = K\omega_n^2 te^{-\omega_n t}u(t)$
$o(t) = K[1 - e^{-\omega_n t} - \omega_n te^{-\omega_n t}]u(t)$
$s_{1,2} = -\omega_n$



3) Underdamped ($0 < \zeta < 1$): Complex pair.
$h(t) = K\frac{\omega_n^2}{\omega_d}e^{-\sigma t}\sin(\omega_d t)u(t)$
$o(t) = K[1 - e^{-\sigma t}(\cos(\omega_d t) + \frac{\sigma}{\omega_d}\sin(\omega_d t))]u(t)$
(decay $\sigma = \omega_n\zeta$, natural $\omega_d = \omega_n\sqrt{1-\zeta^2}$).
$s_{1,2} = -\omega_n\zeta \pm j\omega_n(1-\zeta^2)^{1/2}$

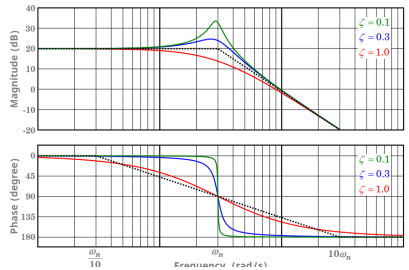
4) Undamped ($\zeta = 0$): Imaginary pair $\pm j\omega_n$.
$s_{1,2} = \pm j\omega_n$



Resonance: Occurs if $\zeta < 1/\sqrt{2}$. ω_r must be real. Resonant Freq $\omega_r = \omega_n\sqrt{1-2\zeta^2}$, Peak$=K/(2\zeta\sqrt{1-\zeta^2})$.

The Bode plots for several values of ζ is shown in Fig.8-1 for the LTI system $\tilde{H}(s) = \frac{10\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$.

Fig.8-1



15. Bode Plots & Asymptotics

- The **Magnitude Plot** is a plot of $\left|\tilde{H}(j\omega)\right|_{dB} = 20\log_{10}\left|\tilde{H}(j\omega)\right|$ dB
- The **Phase Plot** is a plot of $\angle\tilde{H}(j\omega)$ in **degrees**.

8.2 Asymptotic Behavior of Bode Plots

(Assumptions: System is BIBO stable or marginally stable, and has no zero lying on the right-half s-plane)

- **Asymptotic PHASE of Phase Plot** $\left[\angle\tilde{H}(j\omega)\right]$

High frequency:

$$\lim_{\omega\to\infty}\angle\tilde{H}(j\omega) = \overbrace{[\text{№ of Poles} - \text{№ of Zeros}]}^{\text{POLE-ZERO EXCESS}}\times(-90^\circ) \tag{8.4a}$$

Low frequency:

$$\lim_{\omega\to 0}\angle\tilde{H}(j\omega) = \left[\text{№ of }\int dt - \text{№ of }\frac{d}{dt}\right]\times(-90^\circ) \tag{8.4b}$$

- **Asymptotic SLOPE of Magnitude Plot** $\left[\left|\tilde{H}(j\omega)\right|_{dB}\right]$

High frequency:

$$\lim_{\omega\to\infty}\left[\text{slope of }\left|\tilde{H}(j\omega)\right|_{dB}\right] = \overbrace{[\text{№ of Poles} - \text{№ of Zeros}]}^{\text{POLE-ZERO EXCESS}}\times(-20\ dB/decade) \tag{8.5a}$$

Low frequency:

$$\lim_{\omega\to 0}\left[\text{slope of }\left|\tilde{H}(j\omega)\right|_{dB}\right] = \left[\text{№ of }\int dt - \text{№ of }\frac{d}{dt}\right]\times(-20\ dB/decade) \tag{8.5b}$$

$+20L$ dB/decade → Existence of a cascade of L identical differentiators, $(K_d s)^L$. Equation of low frequency asymptote:

$$\left|\tilde{H}(j\omega)\right|_{dB} = 20L\log_{10}(K_d) + 20L\log_{10}(\omega)$$

which cuts the coordinates $(1/K_d,\ 0)$ and $(1,\ 20L\log_{10}(K_d))$.

$-20L$ dB/decade → Existence of a cascade of L identical integrators, $(K_i/s)^L$. Equation of low frequency asymptote:

$$\left|\tilde{H}(j\omega)\right|_{dB} = 20L\log_{10}(K_i) - 20L\log_{10}(\omega)$$

which cuts the coordinates $(K_i,\ 0)$ and $(1,\ 20L\log_{10}(K_i))$.

0 dB/decade → DC gain, K_{dc} (without differentiator or integrator). Equation of low frequency asymptote:

$$\left|\tilde{H}(j\omega)\right|_{dB} = 20\log_{10}(K_{dc}).$$

$+20L$ dB/decade **slope change** at $\omega = \omega_o$ → Existence of a cascade of L identical zero factors, $(s/\omega_o+1)^L$, assuming that all the zeros of $\tilde{H}(s)$ are real.

$-20L$ dB/decade **slope change** at $\omega = \omega_o$ →
- $L = 1$: Existence of a pole factor $\frac{1}{s/\omega_o+1}$.
- $L = 2$: Existence of a repeated pole factor $\left(\frac{1}{s/\omega_o+1}\right)^2$, or a 2nd-order factor $\frac{\omega_o^2}{s^2+2\zeta\omega_o s+\omega_o^2}$
- $L > 2$: May be a hybrid of pole factors and 2nd-order factors, $\left(\frac{1}{s/\omega_o+1}\right)^p\left(\frac{\omega_o^2}{s^2+2\zeta\omega_o s+\omega_o^2}\right)^q$, where $p+2q = L$.

16. DC gain of a Nth-order LTI system:

1) If there are more differentiators, DC gain is $\tilde{H}(0) = 0$.
2) If there are more integrators, DC gain is $\tilde{H}(0) = \infty$.
3) *DC gain should not be expressed in dB, although dB is used for gain.*

19. Resonance in 2nd-order systems

Given $\tilde{H}(s) = \frac{K\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$, resonance happens when $\zeta < 1/\sqrt{2}$.

1) Resonant frequency: $\omega_r = \omega_n\sqrt{1-2\zeta^2}$

2) Resonant peak: $M_r = |\tilde{H}(j\omega_r)| = \frac{K}{2\zeta\sqrt{1-\zeta^2}}$

20. Sampling below Nyquist rate for bandpass signals

1) Perfectly-overlapping images: $f_s = \frac{2f_c}{k}$, where $k = 1, 2, \ldots, \lfloor \frac{2f_c}{B} \rfloor$.

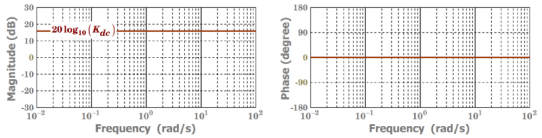
2) Un-aliased spectral images: $\frac{2f_c+B}{k+1} \leqslant f_s \leqslant \frac{2f_c-B}{k}$, $k = 1, 2, \ldots, \lfloor \frac{2f_c-B}{2B} \rfloor$.

3) The above two are only possible when $f_c \geqslant B$.

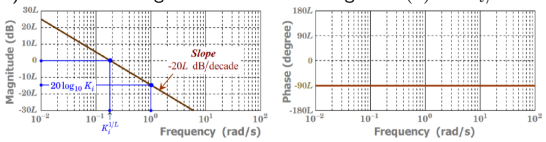
4) Notice: for perfectly-overlapping images, a **division-by-2** is necessary on the reconstruction filter.

21. Drawing of straight-line Bode plots:

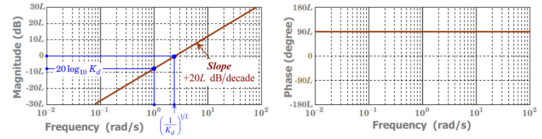
1) Constant $\tilde{H}(s) = K_{dc}$:



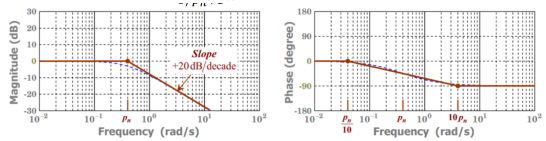
2) L cascaded integrators with combined gain $\tilde{H}(s) = K_i/s^L$:



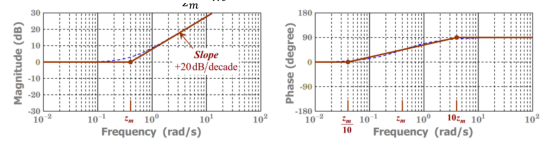
3) L cascaded differentiators with combined gain $\tilde{H}(s) = K_d \cdot s^L$:



4) Pole factor $\tilde{H}(s) = \frac{1}{s/p_n+1}$ with $\tilde{H}(0) = 1$:



5) Zero factor $\tilde{H}(s) = \frac{s}{z_m} + 1$ with $\tilde{H}(0) = 1$:



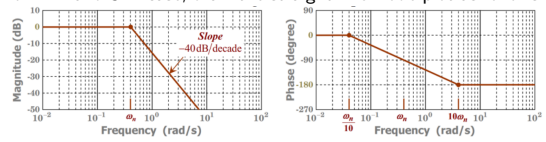
6) 2nd-order factor $\tilde{H}(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$ with $\tilde{H}(0) = 1$:

a) When $\zeta > 1$ (**over-damped**), treat as two cascaded pole factors;

b) When $\zeta = 1$ (**critical-damped**), treat as two repeated pole factors;

c) When $\zeta < 1$ (**under-damped**), approximate as $\zeta = 1$.

For 2nd and 3rd case, draw the straight-line Bode plot as follows:



The rule states that the phase shift caused by a pole or zero is approximated as a straight sloping line that:
Starts exactly **one decade before** the corner frequency. /10
Ends exactly **one decade after** the corner frequency. x10

VALIDITY of $\tilde{H}(j\omega)|_{\omega=2\pi f} = H(f)$

Impulse Response $h(t)$	Transfer Function $\tilde{H}(s) = \mathcal{L}\{h(t)\}$	Stability	Frequency Response $\tilde{H}(j\omega) = \tilde{H}(s)\|_{s=j\omega}$	Frequency Response $H(f) = \mathcal{F}\{h(t)\}$	Validity of $\tilde{H}(j\omega)\|_{\omega=2\pi f} = H(f)$
$u(t)$	$\frac{1}{s}$	Marginally Stable	$\frac{1}{j\omega}$	$\frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$	Everywhere except $\omega = 0$ (or $f = 0$)
$e^{-\alpha t}u(t);\ \alpha > 0$	$\frac{1}{s+\alpha}$	BIBO Stable	$\frac{1}{\alpha + j\omega}$	$\frac{1}{\alpha + j2\pi f}$	Everywhere
$\sin(\omega_o t)u(t)$	$\frac{\omega_o}{s^2+\omega_o^2}$	Marginally Stable	$\frac{\omega_o}{\omega_o^2-\omega^2}$	$\frac{1}{2\pi}\frac{f_o}{f_o^2-f^2} + \frac{j}{4}\left[\begin{matrix}\delta(f+f_o)\\ -\delta(f-f_o)\end{matrix}\right]$	Everywhere except $\omega = \pm\omega_o$ (or $f = \pm f_o$)
$e^{-\alpha t}\sin(\omega_o t)u(t);$ $\alpha > 0$	$\frac{\omega_o}{(s+\alpha)^2+\omega_o^2}$	BIBO Stable	$\frac{\omega_o}{(j\omega+\alpha)^2+\omega_o^2}$	$\frac{2\pi f_o}{(j2\pi f+\alpha)^2+4\pi^2 f_o^2}$	Everywhere
$\cos(\omega_o t)u(t)$	$\frac{s}{s^2+\omega_o^2}$	Marginally Stable	$\frac{j\omega}{\omega_o^2-\omega^2}$	$\frac{1}{2\pi}\frac{jf}{f_o^2-f^2} + \frac{1}{4}\left[\begin{matrix}\delta(f+f_o)\\ +\delta(f-f_o)\end{matrix}\right]$	Everywhere except $\omega = \pm\omega_o$ (or $f = \pm f_o$)
$e^{-\alpha t}\cos(\omega_o t)u(t);$ $\alpha > 0$	$\frac{s+\alpha}{(s+\alpha)^2+\omega_o^2}$	BIBO Stable	$\frac{j\omega+\alpha}{(j\omega+\alpha)^2+\omega_o^2}$	$\frac{j2\pi f+\alpha}{(j2\pi f+\alpha)^2+4\pi^2 f_o^2}$	Everywhere
$te^{-\alpha t}u(t);\ \alpha > 0$	$\frac{1}{(s+\alpha)^2}$	BIBO Stable	$\frac{1}{(\alpha+j\omega)^2}$	$\frac{1}{(\alpha+j2\pi f)^2}$	Everywhere

For checking validity of $H(jw)|_{w=2\pi f} = H(f)$

Transfer Function → Bode Plot (Sketching)

Standardize: Factor TF to isolate K, origin terms $(s,\ 1/s)$, and corner frequencies $(1 + s/\omega_c)$. **Start (Low-Freq):**

- Initial Slope = [#integrators − #differentiators] × (−20 dB/dec).
- Initial Phase = [#integrators − #differentiators] × (−90°).

Navigate Corners (ω_c): Change magnitude slope at each corner.

- Hit a zero: Slope bends UP by $+20L$ dB/dec (L = multiplicity).
- Hit a pole: Slope bends DOWN by $-20L$ dB/dec.

textbfVerify End (High-Freq): Ensure final asymptotes match total system counts.

- Final Slope = [#total poles − #total zeros] × (−20 dB/dec).

Transformation Rules for Functions

Function Notation	Type of Transformation	Change to Coordinate Point
f(x) + d	Vertical translation up d units	(x, y) → (x, y + d)
f(x) − d	Vertical translation down d units	(x, y) → (x, y − d)
f(x + c)	Horizontal translation left c units	(x, y) → (x − c, y)
f(x − c)	Horizontal translation right c units	(x, y) → (x + c, y)
−f(x)	Reflection over x-axis	(x, y) → (x, −y)
f(−x)	Reflection over y-axis	(x, y) → (−x, y)
af(x)	Vertical stretch for |a| > 1	(x, y) → (x, ay)
	Vertical compression for 0 < |a| < 1	
f(bx)	Horizontal compression for |b| > 1	$(x, y) \to \left(\frac{x}{b}, y\right)$
	Horizontal stretch for 0 < |b| < 1	

The product of: Even x Even = Even — Odd x Odd = Even — Odd x Even = Odd — Even/Even = Even — Odd/Odd = Even — Odd/Even = Odd — Even/Odd = Odd

Even: $f(-x) = f(x)$ Odd: $f(-x) = -f(x)$

M% Power Containment Bandwidth (B) The $M\%$ power containment bandwidth is defined as the frequency range containing a specific percentage of the signal's total power or energy.

- **General Discrete Case:** $B = Kf_p$
- K is the smallest positive integer that satisfies:

$$\sum_{k=-K}^{K} |c_k|^2 \geqslant \frac{M}{100} \times P$$

- **Lowpass Signal $x(t)$:** The bandwidth B satisfies:

$$\int_{-B}^{B} E_x(f)df = \frac{M}{100} \times E$$

- **Bandpass Signal $x(t)$:** Centered at f_c, the bandwidth B satisfies:

$$\int_{f_c-0.5B}^{f_c+0.5B} E_x(f)df = \frac{M}{100} \times \frac{E}{2}$$

3dB Bandwidth Definition The 3dB bandwidth of a lowpass signal $x(t)$ is defined as the frequency f_B where $|X(f)|$ first drops to $1/\sqrt{2}$ (≈ -3 dB) of its peak value.

- **Magnitude Criterion:** $|X(f_B)| = \frac{|X(0)|}{\sqrt{2}} = \frac{|X(0)|}{2^{0.5}}$
- **Energy Spectral Density Criterion:** $\frac{E_x(f_B)}{E_x(0)} = \frac{|X(f_B)|^2}{|X(0)|^2} = \frac{1}{2}$

1st-Null Bandwidth

- **Fourier Transform**
- **Null Frequencies:** Occur where the argument of the sinc function is an integer: $f = \pm 4, \pm 8, \pm 12 \ldots$ Hz.
- **Result:** The 1st-null bandwidth is 4 Hz.

Fourier Duality & Properties

- **Periodic in Time ↔ Discrete in Frequency:** If $x(t)$ has period T, spectrum consists of impulses at k/T.
- **Discrete in Time ↔ Periodic in Frequency:** Sampling a signal makes its spectrum periodic.

Fourier Series Representations

- **Complex Exponential Fourier Series**
  - *Fourier Analysis (Forward):*

$$c_k = \frac{1}{T_p}\int_{t_0}^{t_0+T_p} x_p(t)e^{-j2\pi kt/T_p}dt; \quad \forall \text{ integer}$$

  - *Fourier Synthesis (Inverse):*

$$x_p(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi kt/T_p}$$

- **Trigonometric Fourier Series**
  - *Fourier Analysis (Forward):*

$$a_k = \frac{1}{T_p}\int_{t_0}^{t_0+T_p} x_p(t)\cos(2\pi kt/T_p)dt; \quad k \geqslant 0$$

$$b_k = \frac{1}{T_p}\int_{t_0}^{t_0+T_p} x_p(t)\sin(2\pi kt/T_p)dt; \quad k > 0$$

  - *Fourier Synthesis (Inverse):*

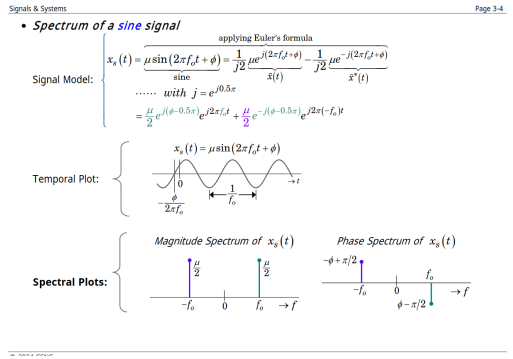
$$x_p(t) = a_0 + 2\sum_{k=1}^{\infty}\left[a_k\cos(2\pi kt/T_p) + b_k\sin(2\pi\right.$$

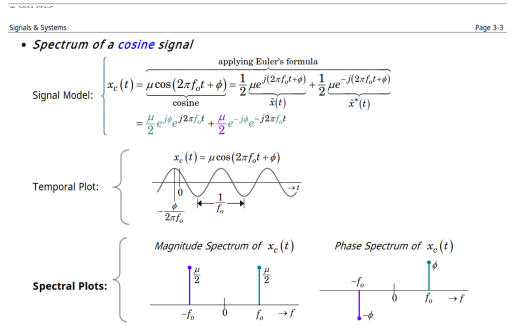
The DC Component (Zero Frequency)

Perspective	Formula/Coefficient
Physical	Average value over one period.
Complex (c_k)	c_0
Trig (a_n, b_n)	$a_0/2$
Visual	The "center lineör vertical offset of the wave.

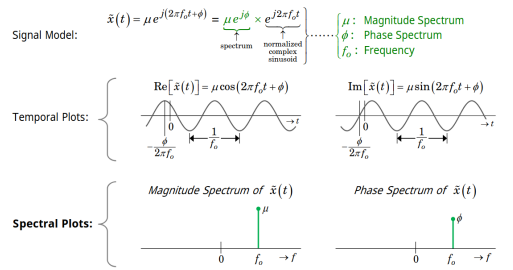
Signal Properties and Periodicity The inverse transform is used to determine signal characteristics:

- **Real Signals:** Convert complex exponentials to sin/cosine form. If any imaginary components remain, the signal is not purely real.
- **Periodicity:** A signal is periodic if there is a common fundamental frequency (HCF) among all components.
  - The ratio of any two frequencies in the signal must be a **rational number** (a fraction of two integers).
  - A signal becomes **completely aperiodic** when the ratio of its frequencies is **irrational**.
  - **Note:** The HCF can be a decimal (e.g., 0.1 or 0.01), but it cannot be an irrational constant like π or $\sqrt{2}$.





3.2 Spectrum of a Sinusoid



Sampling using f_s, the magnitude of the sampled signal is SCALED by the sampling frequency. This requires the ideal filter to scale down later.